

MODULE 46 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 2

Fifteen questions on designing new solutions. The heaviest domain, at 29%.

DOMAIN 2 of 4 EXAM WEIGHT 29% SET 2 of 5

A shared networking stack creates a VPC and subnets. Thirty application stacks, owned by different teams, need the VPC and subnet IDs. Teams deploy on their own schedules and must not be blocked by, or able to affect, the networking stack's deployments.

The networking team updates its stack roughly monthly.

How should the application stacks obtain the IDs?

- ☐ A Make the application stacks nested stacks of the networking stack
- ☐ B Export the values from the networking stack and import them with `Fn::ImportValue`
- ☐ C Hard-code the IDs in each application template
- ☐ D Pass the IDs as parameters supplied manually at each deployment

◆ PAUSE AND DECIDE

How should the application stacks obtain the IDs?

A Nested stacks

Nested stacks share one lifecycle, so every application deploys when the parent deploys. That is precisely the coupling the requirement forbids.

C Hard-code

Thirty templates to edit whenever anything changes, and no relationship CloudFormation can protect.

B Export and ImportValue

Independent lifecycles with a real dependency. Note the trade: an exported value cannot be changed while it is imported.

D Manual parameters

Works and depends on a human supplying the right value thirty times. Nothing prevents a wrong one.

THE SIGNAL

"Deploy on their own schedules" is the divider. Nested means one lifecycle; exports mean many. Module 12.

A new ECS service on Fargate must deploy with zero downtime, shift traffic gradually, and roll back automatically within two minutes if the 5xx rate rises. The team wants to avoid managing additional deployment tooling.

The service sits behind an Application Load Balancer.

What should the solutions architect recommend?

- ☐ A Rolling update with a longer health check grace period
- ☐ B Blue/green through CodeDeploy, with a manual rollback runbook
- ☐ C ECS built-in blue/green deployments with a CloudWatch alarm configured as a rollback trigger
- ☐ D Deploy to a second cluster and switch with a weighted Route 53 record

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Rolling update
Replaces tasks in place, so a bad version is partially live and rollback means another rolling deployment.
- B

 CodeDeploy with a manual runbook
Manual rollback cannot meet two minutes reliably, and it adds the extra tooling the team wanted to avoid.
- C

 ECS native blue/green with alarm rollback
Native since 2025, with canary and linear shifting, and an alarm trigger reverses automatically. AWS now recommends it for new ECS deployments.
- D

 Second cluster plus DNS
DNS caching means you cannot guarantee when a shift or a rollback takes effect.

THE SIGNAL

Automatic rollback within a stated time always means an alarm trigger, never a runbook. Module 13.

A nightly genomics pipeline processes 40,000 independent tasks. Individual task failures can be retried without consequence, and the pipeline has a twelve-hour window to complete a four-hour workload. Cost is the primary concern.

The workload runs on Linux and is architecture-agnostic.

What should the solutions architect recommend?

- ☐ A An EC2 Auto Scaling group of Spot Instances using the price-capacity-optimized allocation strategy, across several instance types
- ☐ B On-Demand Instances with a three-year Compute Savings Plan
- ☐ C Spot Fleet with the lowest-price allocation strategy
- ☐ D Reserved Instances sized to peak nightly demand

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Spot via ASG, price-capacity-optimized
Fault-tolerant, flexible timing, architecture-agnostic: the Spot profile exactly. Price-capacity-optimized is AWS's recommended strategy, and Spot Fleet is legacy.
- B** On-Demand with a Savings Plan
Commits for three years to a workload that runs four hours a night and tolerates interruption. Spot is far cheaper here.
- C** Spot Fleet, lowest-price
Two dated choices. AWS labels Spot Fleet legacy and advises against lowest-price, which chases cheap pools with poor capacity.
- D** Reserved Instances
Pays for capacity around the clock for a workload using it four hours a night.

THE SIGNAL

Option C is right about Spot and wrong on both details. The strategy and the service have both moved on. Module 14.

A company is designing a new container platform. It has four years of Helm charts, custom operators and Kubernetes-native tooling, and its platform team is deeply skilled in Kubernetes. A consultant recommends ECS on the grounds that it has lower operational overhead.

The workloads are standard long-running services.

What should the solutions architect advise?

- ☐ A Adopt ECS, since lower operational overhead is the deciding factor
- ☐ B Adopt ECS but keep Kubernetes on-premises for the operators
- ☐ C Adopt App Runner, which removes container orchestration entirely
- ☐ D Adopt EKS, because the existing Kubernetes investment and skills outweigh ECS's operational simplicity

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Adopt ECS

True in general and wrong here. Helm charts and custom operators do not run on ECS, so the whole investment is discarded.

C App Runner

Also in maintenance since March 2026, and it cannot run custom operators at all.

B ECS plus on-premises Kubernetes

Two platforms, two operational models, and a hybrid estate created deliberately to avoid using EKS.

D Adopt EKS

Existing artefacts and skills change the answer. Four years of Helm charts and operators transfer to EKS unchanged.

THE SIGNAL

ECS is simpler in the abstract. Existing Kubernetes artefacts and skills are the standard exception, and they appear often. Module 15.

A new application on Windows Server instances requires a shared file system accessible over SMB, with Active Directory integration for per-user access control and Windows ACL support. It must be available across two Availability Zones. Throughput requirements are modest.

Which storage service should be used?

- ☐ A Amazon EFS with a mount target in each AZ
- ☐ B Amazon FSx for Windows File Server, Multi-AZ
- ☐ C Amazon FSx for Lustre with an S3 data repository
- ☐ D An EBS Multi-Attach volume shared between instances

◆ PAUSE AND DECIDE

Which storage service should be used?

- A

 EFS
EFS speaks NFS, not SMB, and does not provide Windows ACLs or native AD integration.
- B

 FSx for Windows File Server
Native SMB, Windows ACLs and AD integration, with a Multi-AZ option. Every stated requirement maps directly.
- C

 FSx for Lustre
A high-performance parallel file system for HPC and analytics. Wrong protocol and wrong problem.
- D

 EBS Multi-Attach
Attaches one volume to several instances in one AZ, and needs a cluster-aware file system. It is not a shared file service.

THE SIGNAL

SMB plus Active Directory plus Windows ACLs is the FSx for Windows signature, and all three appear together. Module 16.

A new mobile backend needs single-digit millisecond reads on key lookups, unpredictable traffic that may go from zero to 50,000 requests per second within minutes, and no capacity planning or patching. The access pattern is a single lookup by user ID.

The team has no database administrators.

What should the solutions architect recommend?

- ☐ A Amazon RDS for PostgreSQL with read replicas
- ☐ B Aurora Serverless with a read-only endpoint
- ☐ C Amazon DynamoDB in on-demand capacity mode
- ☐ D Amazon ElastiCache for Valkey in front of RDS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

RDS with read replicas
Relational capability the access pattern does not need, plus instance sizing and replica management for a team with no DBAs.
- B

Aurora Serverless
The strongest distractor. It scales and it is still a relational engine sized in ACUs, for a pure key lookup.
- C

DynamoDB on-demand
Single-key access, single-digit millisecond reads, instant scaling with no capacity planning, nothing to patch. On-demand is now AWS's recommended default.
- D

ElastiCache plus RDS
Two systems to operate and a cache-coherence problem, to serve an access pattern DynamoDB handles natively.

THE SIGNAL

"A single lookup by user ID" tells you no relational capability is needed. That sentence is the whole question. Module 17.

An order system must process events for each customer strictly in the order they were produced, and must never process the same event twice. Events for different customers may be processed concurrently, and throughput is around 2,000 messages per second. Consumers are Lambda functions.

What should the solutions architect recommend?

- ☐ A An SQS FIFO queue, using the customer ID as the message group ID
- ☐ B An SQS standard queue with a deduplication table in DynamoDB
- ☐ C An SNS standard topic fanning out to Lambda
- ☐ D An EventBridge bus with a rule per customer

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 FIFO with customer ID as group ID
Ordering and exactly-once are guaranteed within a message group, and separate groups process in parallel. Both requirements, one setting.
- B

 Standard queue plus dedupe table
Handles duplicates and not ordering, and builds a mechanism FIFO provides natively.
- C

 SNS standard topic
SNS offers neither ordering nor exactly-once delivery in its standard form.
- D

 EventBridge rule per customer
EventBridge does not guarantee ordering, and a rule per customer does not scale to a real customer base.

THE SIGNAL

The message group ID is what makes per-customer ordering and concurrency compatible. Without it, FIFO serialises everything. Module 18.

A new payments platform requires an RTO of 15 minutes and an RPO of 5 minutes in a second Region. The business will fund standing infrastructure but wants to avoid running full production capacity twice. Failover must not depend on launching large numbers of instances during a Regional event.

The database is Aurora.

Which strategy fits?

- ☐ A Backup and restore, with cross-Region snapshot copies
- ☐ B Pilot light: database replicating, compute defined but not running
- ☐ C Multi-site active/active across both Regions
- ☐ D Warm standby: a scaled-down but running copy, with Aurora Global Database

◆ PAUSE AND DECIDE

Which strategy fits?

A Backup and restore

Hours of RTO. It is an order of magnitude away from fifteen minutes.

C Multi-site active/active

Meets the objectives and means running full production capacity twice, which the business explicitly declined to fund.

B Pilot light

The strong near miss. RPO is fine, and recovery depends on launching compute during the event, which the scenario rules out.

D Warm standby with Aurora Global

Something is always running, so failover means scaling up rather than creating. Aurora Global Database gives roughly one-second RPO.

THE SIGNAL

"Must not depend on launching instances during the event" is a static-stability requirement, and it eliminates pilot light. Modules 19 and 30.

A new application connects to Amazon RDS for PostgreSQL. Security requires that the database password rotates every 30 days, that no developer ever sees it, and that rotation does not require an application deployment or restart.

The application is written in Python.

What should the solutions architect recommend?

- ☐ A Store the password in Parameter Store as a SecureString and rotate it with a scheduled Lambda
- ☐ B Use AWS Secrets Manager with managed rotation for RDS, and retrieve the secret at runtime
- ☐ C Store the password in an encrypted environment variable, redeployed monthly
- ☐ D Use IAM database authentication with a token refreshed every 15 minutes

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

Parameter Store plus custom Lambda
Parameter Store has no built-in rotation, so you write and maintain the rotation logic Secrets Manager provides.
- B

Secrets Manager with managed RDS rotation
Managed rotation changes the password in the database and the secret together. Runtime retrieval means no redeploy and no developer ever sees it.
- C

Environment variable
Requires a deployment to rotate, which the requirement forbids, and the value is visible to anyone who can read the configuration.
- D

IAM database authentication
A genuinely strong option that eliminates the password entirely, and it has connection-rate limits and changes how the application authenticates.

THE SIGNAL

Rotation without redeployment is the divider between Secrets Manager and Parameter Store. One has it built in. Module 20.

A new multiplayer game server uses UDP. Players worldwide report inconsistent latency and long recovery when a Regional endpoint degrades. The company needs static entry-point IP addresses for allowlisting on corporate networks, and failover in seconds. Content is dynamic and not cacheable.

What should the solutions architect recommend?

- ☐ A Amazon CloudFront with a custom origin
- ☐ B Route 53 latency-based routing with health checks
- ☐ C AWS Global Accelerator with endpoint groups in each Region
- ☐ D An Application Load Balancer with cross-zone load balancing

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 CloudFront
Caches HTTP content. It does not carry UDP, and the scenario says the content is not cacheable anyway.
- B

 Route 53 latency routing
The near miss. It directs to the nearest healthy Region, and failover waits for DNS TTLs rather than taking seconds.
- C

 Global Accelerator
Any protocol including UDP, two static anycast IPs for allowlisting, traffic on the AWS backbone, and failover in seconds without DNS.
- D

 ALB with cross-zone
Regional and HTTP-only. It addresses neither the global latency nor the UDP requirement.

THE SIGNAL

Three signals point the same way: UDP, static IPs, and seconds. Any one of them rules CloudFront and DNS failover out. Module 21.

A new data pipeline in private subnets will read and write roughly 90 TB per month to Amazon S3 in the same Region. The architecture routes all outbound traffic through NAT gateways. A cost review flags NAT data processing charges as the largest line item in the design.

The pipeline also calls a small number of third-party APIs.

What is the MOST effective change?

- ☐ A Add a gateway VPC endpoint for S3, so that S3 traffic bypasses the NAT gateways entirely
- ☐ B Add an interface VPC endpoint for S3
- ☐ C Move the pipeline into public subnets with public IP addresses
- ☐ D Compress objects before writing them to S3

◆ PAUSE AND DECIDE

What is the MOST effective change?

- A

 Gateway endpoint for S3
Gateway endpoints cost nothing and remove 90 TB a month from NAT data processing. The third-party calls still use the NAT.
- B

 Interface endpoint for S3
Also bypasses the NAT, and charges hourly plus per GB. A gateway endpoint does the same job for free.
- C

 Public subnets
Removes NAT charges by removing network isolation. It trades a cost problem for a security one.
- D

 Compression
Worth doing and it reduces the volume rather than eliminating the charge. Gateway endpoints remove it entirely.

THE SIGNAL

Gateway endpoints are free; interface endpoints are not. For S3 and DynamoDB in-Region, that is the whole comparison. Modules 09 and 22.

A CloudFormation template creates a production RDS instance. The team must ensure the database survives two situations: the stack being deleted, and a template change that causes CloudFormation to replace the resource rather than update it in place. A previous incident lost data during a stack update.

What should the template specify?

- ☐ A DeletionPolicy set to Snapshot
- ☐ B UpdateReplacePolicy set to Retain
- ☐ C A stack policy denying updates to the database resource
- ☐ D Both DeletionPolicy and UpdateReplacePolicy, since each covers a different event

◆ PAUSE AND DECIDE

What should the template specify?

- A

 DeletionPolicy only
Covers stack deletion. A replacement during an update is a different event and is not covered.
- B

 UpdateReplacePolicy only
Covers replacement during update. Deleting the stack would still remove the database.
- C

 Stack policy
Prevents the update rather than protecting the data, and blocks legitimate changes. It also does nothing on stack deletion.
- D

 Both policies
Two events, two attributes. The previous incident was a replacement during update, which is exactly the half people forget.

THE SIGNAL

The scenario names both events deliberately. Setting only DeletionPolicy is the commonest production mistake here. Module 12.

A new financial application needs a relational database with strongly consistent reads and transactions in its primary Region, low-latency reads for users in three other Regions, and a cross-Region RPO of about one second with failover in under a minute.

The schema is relational with foreign keys.

What should the solutions architect recommend?

- ☐ A DynamoDB global tables with strongly consistent reads
- ☐ B Aurora Global Database with secondary Regions for reads
- ☐ C RDS Multi-AZ with cross-Region read replicas
- ☐ D Aurora DSQL for active-active multi-Region writes

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

DynamoDB global tables
Not relational, no foreign keys or joins, and global tables replicate with eventual consistency between Regions.
- B

Aurora Global Database
Relational with transactions, sub-second cross-Region replication, secondary Regions serve local reads, and failover typically under a minute.
- C

RDS cross-Region replicas
Replication lag is higher and less predictable, and promotion is slower than Aurora Global's managed failover.
- D

Aurora DSQL
Genuinely impressive for active-active, and it does not support foreign keys, and it is not on the exam's in-scope list.

THE SIGNAL

"Relational with foreign keys" rules out both DynamoDB and DSQL in one phrase. Module 17.

A new account will run a critical payment Lambda function alongside a batch reporting function that occasionally scales to thousands of concurrent executions. The payment function must always have capacity available, and the batch function must never be able to consume the whole account's concurrency.

Cold starts are not a stated concern.

What should the solutions architect configure?

- ☐ A Provisioned concurrency on the payment function
- ☐ B A concurrency quota increase for the account
- ☐ C Reserved concurrency on both functions
- ☐ D SnapStart on the payment function

◆ PAUSE AND DECIDE

What should the solutions architect configure?

- A

 Provisioned concurrency
Solves cold-start latency, which the scenario explicitly sets aside. It does not stop the batch function consuming the pool.
- B

 Quota increase
A bigger shared pool that the batch function can still exhaust. It raises the ceiling without partitioning it.
- C

 Reserved concurrency on both
Reserved is a ceiling and a floor at once: it caps the batch function and guarantees the payment function its share.
- D

 SnapStart
Also a cold-start feature, and it is limited to Java, Python and .NET. No runtime is stated.

THE SIGNAL

"Cold starts are not a stated concern" is planted to eliminate the two latency answers. Reserved is the contention answer. Module 15.

A company runs its production ERP on-premises and intends to keep it there. It needs a disaster recovery capability in AWS with an RPO measured in seconds and an RTO under 20 minutes, testable without disrupting production.

A colleague has suggested AWS Application Migration Service.

What should the solutions architect advise?

- ☐ A Use AWS Elastic Disaster Recovery, since production stays on-premises and this is DR rather than migration
- ☐ B Use Application Migration Service as suggested, since the replication is identical
- ☐ C Use AWS Backup with cross-Region copies
- ☐ D Rehost the ERP in AWS, then use Multi-AZ for resilience

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Elastic Disaster Recovery
DRS protects production that stays put, giving RPO in seconds and RTO of 5-20 minutes, with non-disruptive drills. MGN migrates permanently.
- B

 MGN as suggested
The replication technology is largely shared, and MGN is built for a one-way permanent move, not ongoing protection and repeated failback.
- C

 AWS Backup
Hours of RPO and RTO. It is an order of magnitude away from seconds and twenty minutes.
- D

 Rehost the ERP
The scenario states production stays on-premises. This answers a question nobody asked.

THE SIGNAL

MGN and DRS share most of their technology and answer opposite questions. Ask whether production is moving or staying. Modules 19 and 35.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q12	Module 12 – Infrastructure as code
Q2	Module 13 – Deployment strategies and rollback
Q3, Q4, Q14	Modules 14-15 – Compute, containers and serverless
Q5, Q6, Q13	Modules 16-17 – Storage and databases
Q7, Q8, Q15	Modules 18-19 – Integration and continuity
Q9, Q10, Q11	Modules 20-22 – Security, performance and cost

ON THE EXAM

Domain 2 is 29% of the exam, the largest single block. 12 or more out of 15 is on track; below 9, rewatch before continuing.